



Question 1) After passing through a Stern-Gerlach magnet oriented along an arbitrary axis $\mathbf{n} = (\sin \beta \cos \alpha, \sin \beta \sin \alpha, \cos \beta)$, the beam is split into two spin states:

$$\begin{aligned} |S_{\mathbf{n}}; +\rangle &= \cos \frac{\beta}{2} |+\rangle + e^{i\alpha} \sin \frac{\beta}{2} |-\rangle, \\ |S_{\mathbf{n}}; -\rangle &= -e^{-i\alpha} \sin \frac{\beta}{2} |+\rangle + \cos \frac{\beta}{2} |-\rangle. \end{aligned} \quad (1)$$

a) Find these states by diagonalizing the spin operator

$$\begin{aligned} \hat{S}_{\mathbf{n}} &= n_x \hat{S}_x + n_y \hat{S}_y + n_z \hat{S}_z \\ &= \frac{\hbar}{2} (\cos \beta |+\rangle \langle +| + e^{-i\alpha} \sin \beta |+\rangle \langle -| + e^{i\alpha} \sin \beta |-\rangle \langle +| - \cos \beta |-\rangle \langle -|) \end{aligned} \quad (2)$$

$\hat{S}_{\mathbf{n}}$ along the axis $\mathbf{n}$. That is, treat this as an eigenvalue problem (similarly to what S&N tells us to do in **Problem 1.11**). Feel free to use the matrix representation for our states

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (3)$$

and whatever method you prefer.

b) Calculate the inner product $\langle S_{\mathbf{n}}; + | \hat{S}_{\mathbf{n}} | + \rangle$. Are the states $|S_{\mathbf{n}}; +\rangle$ and $\hat{S}_{\mathbf{n}} |+\rangle$ the same?

c) In what direction does the Bloch vector of a Stern-Gerlach magnet point for the states $|S_{\mathbf{n}}; +\rangle$ and $|S_{\mathbf{n}}; -\rangle$? Recall that the Bloch vector for a state $|\psi\rangle$ is given by $\langle \hat{\mathbf{S}} \rangle = \langle \psi | \hat{\mathbf{S}} | \psi \rangle$.

d) Each Stern-Gerlach magnet can be described by a projector operator

$$\hat{P}_{\mathbf{n}; \pm} = |S_{\mathbf{n}}; \pm\rangle \langle S_{\mathbf{n}}; \pm|. \quad (4)$$

After passing through a magnet oriented in the direction $\mathbf{n}$ that blocks $\mp$, but allows $\pm$, we have a pure state

$$|S_{\mathbf{n}}; \pm\rangle. \quad (5)$$

We do not ask about what happened earlier than this point, all we care about is that this new state was created. Suppose now we add a second magnet oriented in the direction $\mathbf{m}$. After exiting through the second magnet, the probability of finding the particle in the state $|S_{\mathbf{m}}; \pm\rangle$ is given by

$$\langle S_{\mathbf{n}}; \pm | \hat{P}_{\mathbf{m}; \pm} | S_{\mathbf{n}}; \pm \rangle = |\langle S_{\mathbf{m}}; \pm | S_{\mathbf{n}}; \pm \rangle|^2. \quad (6)$$

You can think of this as the probability or the relative intensity of the beam. Imagine we add a third magnet oriented in the direction $\mathbf{l}$. What is the probability of finding the particle in the state $|S_{\mathbf{l}}; \pm\rangle$ after passing through all three magnets? Express your answer in terms of the angle $\theta_{\mathbf{lm}}$ between the directions $\mathbf{l}$ and $\mathbf{n}$ and the angle $\theta_{\mathbf{mn}}$ between the directions $\mathbf{m}$ and $\mathbf{n}$.

Hint: you may want to start by proving that the projector $\hat{P}_{\mathbf{n}; \pm}$ is given by

$$\hat{P}_{\mathbf{n}; \pm} = \frac{1 + \mathbf{n} \cdot \hat{\mathbf{S}} / |\mathbf{S}|}{2} \quad (7)$$

e) Calculate the probability of finding the particle in the state $|S_{\mathbf{l}}; \pm\rangle$ after passing through all three magnets for the following arrangements:

$$(a) \quad \mathbf{n} = \hat{\mathbf{z}}, \quad \mathbf{m} = -\hat{\mathbf{z}}, \quad \mathbf{l} = \hat{\mathbf{z}}$$

$$(b) \quad \mathbf{n} = \hat{\mathbf{z}}, \quad \mathbf{m} = \hat{\mathbf{x}}, \quad \mathbf{l} = \hat{\mathbf{z}}$$

$$(c) \quad \mathbf{n} = \hat{\mathbf{z}}, \quad \mathbf{m} = (\hat{\mathbf{x}} + \hat{\mathbf{z}})/\sqrt{2}, \quad \mathbf{l} = \hat{\mathbf{x}}$$

- g) Generalize your result for an arbitrary number N of sequential Stern-Gerlach magnets going from an initial $\hat{\mathbf{z}}$ direction to a final $\hat{\mathbf{x}}$ direction in equal angular steps confined to the xz -plane. What happens in the limit $N \rightarrow \infty$? What is the probability of measuring the particle in the state $|S_{\mathbf{x}}; \pm\rangle$? How about $|S_{\mathbf{z}}; \pm\rangle$? How does this compare with case (b) above?

a) In the $\{|+\rangle, |-\rangle\}$ basis the dyadic form given in the problem reads

$$\hat{S}_{\mathbf{n}} = \frac{\hbar}{2} \begin{pmatrix} \cos \beta & e^{-i\alpha} \sin \beta \\ e^{i\alpha} \sin \beta & -\cos \beta \end{pmatrix}. \quad (8)$$

Note:

$$\text{Tr}(\hat{S}_{\mathbf{n}}) = \lambda_1 + \lambda_2 = 0, \quad \det(\hat{S}_{\mathbf{n}}) = \lambda_1 \lambda_2 = -\frac{\hbar^2}{4} (\cos^2 \beta + \sin^2 \beta) = -\frac{\hbar^2}{4} \quad (9)$$

You should know:

- The trace of a matrix is the sum of its eigenvalues:

$$\text{Tr}(M) = \text{Tr}(P\Lambda P^{-1}) = \text{Tr}(\Lambda P^{-1}P) = \text{Tr}(\Lambda) = \sum_i \lambda_i \quad (10)$$

where P is the invertible matrix ($P^{-1}P = \mathbb{1}$) that diagonalizes M , i.e., $P^{-1}MP = \Lambda$, with $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots)$. Note we used the cyclic property of the trace: $\text{Tr}(AB) = \text{Tr}(BA)$.

- The determinant of a matrix is the product of its eigenvalues:

$$\det(M) = \det(P\Lambda P^{-1}) = \det(\Lambda) = \prod_i \lambda_i \quad (11)$$

where we used

$$\det(P\Lambda P^{-1}) = \det(P) \det(\Lambda) \det(P^{-1}) = \det(P P^{-1}) \det(\Lambda) = \det(\Lambda). \quad (12)$$

From these properties, we know: $\lambda_1 = -\lambda_2$ and $\lambda_1 \lambda_2 = -\lambda_1^2 = -\frac{\hbar^2}{4}$, which immediately gives the eigenvalues:

$$\lambda_{1,2} = \pm \frac{\hbar}{2}. \quad (13)$$

Note this is independent of α and β . It had to be the case since the magnet in isolation cannot distinguish any particular direction in space.

Now onto the eigenvectors. Starting with $\lambda_1 = +\hbar/2$, we map the dyad form to matrices:

$$|v_1\rangle \mapsto \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \hat{S}_{\mathbf{n}} \mapsto \frac{\hbar}{2} \begin{pmatrix} \cos \beta & e^{-i\alpha} \sin \beta \\ e^{i\alpha} \sin \beta & -\cos \beta \end{pmatrix} \quad (14)$$

with complex numbers x_1 and y_1 . The eigenvalue problem is just the statement:

$$(\hat{S}_{\mathbf{n}} - \lambda_1 \mathbb{1}) |v_1\rangle = 0 \mapsto \begin{pmatrix} \cos \beta - 1 & e^{-i\alpha} \sin \beta \\ e^{i\alpha} \sin \beta & -\cos \beta - 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = 0. \quad (15)$$

With the help of the identities $\cos \beta - 1 = -2 \sin^2 \frac{\beta}{2}$ and $\sin \beta = 2 \sin \frac{\beta}{2} \cos \frac{\beta}{2}$, you can show that

$$y_1 = e^{i\alpha} \tan \frac{\beta}{2} x_1. \quad (16)$$

Choosing $x_1 = \cos \frac{\beta}{2}$, we normalize the state and make the first coefficient real and non-negative (again the freedom of the global phase), giving our solution

$$|S_{\mathbf{n}}; +\rangle = \cos \frac{\beta}{2} |+\rangle + e^{i\alpha} \sin \frac{\beta}{2} |-\rangle, \quad (17)$$

The same steps for $\lambda = -\hbar/2$ give

$$|S_{\mathbf{n}}; -\rangle = -e^{-i\alpha} \sin \frac{\beta}{2} |+\rangle + \cos \frac{\beta}{2} |-\rangle. \quad (18)$$

Note they are orthogonal,

$$\langle S_{\mathbf{n}}; + | S_{\mathbf{n}}; - \rangle = -e^{-i\alpha} \cos \frac{\beta}{2} \sin \frac{\beta}{2} + e^{-i\alpha} \sin \frac{\beta}{2} \cos \frac{\beta}{2} = 0, \quad (19)$$

as they must be for distinct eigenvalues of a Hermitian operator (your homework 1).

b) Since $\hat{S}_{\mathbf{n}}$ is Hermitian with real eigenvalue $\hbar/2$, it may act to the *left* on its own eigenbra, $\langle S_{\mathbf{n}}; + | \hat{S}_{\mathbf{n}} = \frac{\hbar}{2} \langle S_{\mathbf{n}}; + |$, so

$$\langle S_{\mathbf{n}}; + | \hat{S}_{\mathbf{n}} | + \rangle = \frac{\hbar}{2} \langle S_{\mathbf{n}}; + | + \rangle = \frac{\hbar}{2} \cos \frac{\beta}{2}. \quad (20)$$

They are clearly not the same state unless $\beta = 0$. You can also write the state explicitly to see:

$$\hat{S}_{\mathbf{n}} | + \rangle = \frac{\hbar}{2} (\cos \beta | + \rangle + e^{i\alpha} \sin \beta | - \rangle), \quad (21)$$

This depends on the full angle β rather than $\beta/2$, and is not even normalized.

c) Write $|S_{\mathbf{n}}; +\rangle = a |+\rangle + b |-\rangle$ with $a = \cos \frac{\beta}{2}$ and $b = e^{i\alpha} \sin \frac{\beta}{2}$. Then

$$\langle \hat{S}_x \rangle = \hbar \operatorname{Re}(a^* b), \quad \langle \hat{S}_y \rangle = \hbar \operatorname{Im}(a^* b), \quad \langle \hat{S}_z \rangle = \frac{\hbar}{2} (|a|^2 - |b|^2), \quad (22)$$

and using $a^* b = \cos \frac{\beta}{2} \sin \frac{\beta}{2} e^{i\alpha} = \frac{1}{2} \sin \beta e^{i\alpha}$ and $|a|^2 - |b|^2 = \cos \beta$, you'd get

$$\langle \hat{\mathbf{S}} \rangle_{S_{\mathbf{n}}; +} = \langle S_{\mathbf{n}}; + | \hat{\mathbf{S}} | S_{\mathbf{n}}; + \rangle = \frac{\hbar}{2} (\sin \beta \cos \alpha, \sin \beta \sin \alpha, \cos \beta) = \frac{\hbar}{2} \mathbf{n} \quad (23)$$

For $\langle \hat{\mathbf{S}} \rangle_{S_{\mathbf{n}}; -} = -\frac{\hbar}{2} \mathbf{n}$. So the state leaving the magnet in the eigenstate $|S_{\mathbf{n}}; +\rangle$ has its Bloch vector pointing exactly along the magnet, and reversely for the state leaving the magnet in the eigenstate $|S_{\mathbf{n}}; -\rangle$. The two states are said to be “antipodal” on the Bloch sphere.

d) First the hint. You can rewrite it slightly differently as

$$\hat{P}_{\mathbf{n}; \pm} = |S_{\mathbf{n}}; \pm\rangle \langle S_{\mathbf{n}}; \pm| = \frac{\mathbb{1} \pm \mathbf{n} \cdot \vec{\sigma}}{2}. \quad (24)$$

 You can check $\hat{P}_{\mathbf{n}; \pm}$ follows all the properties of a projector. For example, using the property of Pauli matrices $(\mathbf{n} \cdot \vec{\sigma})^2 = \mathbb{1}$, you can show

$$\hat{P}_{\mathbf{n}; \pm}^2 = \left(\frac{\mathbb{1} \pm \mathbf{n} \cdot \vec{\sigma}}{2} \right)^2 = \frac{\mathbb{1} \pm 2\mathbf{n} \cdot \vec{\sigma} + \mathbb{1}}{4} = \frac{\mathbb{1} \pm \mathbf{n} \cdot \vec{\sigma}}{2} = \hat{P}_{\mathbf{n}; \pm} \quad (25)$$

Also, $\hat{P}_{\mathbf{n}; +} + \hat{P}_{\mathbf{n}; -} = \mathbb{1}$ and $\hat{P}_{\mathbf{n}; +} \hat{P}_{\mathbf{n}; -} = \frac{1}{4} (\mathbb{1} - (\mathbf{n} \cdot \vec{\sigma})^2) = 0$. Finally, note that $\hat{S}_{\mathbf{n}} \hat{P}_{\mathbf{n}; \pm} = \pm \frac{\hbar}{2} \hat{P}_{\mathbf{n}; \pm}$ so we can easily find onto what eigenstate of $\hat{S}_{\mathbf{n}}$ the projector $\hat{P}_{\mathbf{n}; \pm}$ projects.

Equipped with this, we can prove the overlap quite easily. Eq. (24) and then Eq. (23),

$$|\langle S_{\mathbf{a}}; + | S_{\mathbf{b}}; + \rangle|^2 = \langle S_{\mathbf{b}}; + | \hat{P}_{\mathbf{a};+} | S_{\mathbf{b}}; + \rangle = \frac{1 + \frac{2}{\hbar} \mathbf{a} \cdot \langle \hat{\mathbf{S}} \rangle_{\mathbf{b}}}{2} = \frac{1 + \mathbf{a} \cdot \mathbf{b}}{2} = \cos^2 \left(\frac{\theta_{ab}}{2} \right). \quad (26)$$

Each blocked magnet projects and renormalizes the state, so that the survival probabilities multiply. For the chain $\mathbf{n} \rightarrow \mathbf{m} \rightarrow \mathbf{l}$, taking the +ve eigenstate throughout,

$$p = |\langle S_{\mathbf{l}}; + | S_{\mathbf{m}}; + \rangle|^2 |\langle S_{\mathbf{m}}; + | S_{\mathbf{n}}; + \rangle|^2 = \cos^2 \left(\frac{\theta_{lm}}{2} \right) \cos^2 \left(\frac{\theta_{mn}}{2} \right). \quad (27)$$



Note only the angles between consecutive magnets appear. The angle θ_{ln} between the first and last magnets does not appear anywhere. This is natural with the interpretation of each magnet as a projector. Once the atom has been projected onto $|S_{\mathbf{m}}; +\rangle$, it is now in the state $|S_{\mathbf{m}}; +\rangle$ and has no memory of its previous state. Note also that projectors are not reversible, in the same way that measurements in quantum mechanics “collapse the state”.

e) Cases:

1. (a) Using the formula above (Eq. (27)) with $\theta_{-zz} = \pi$ gives $p = 0$. The beam is completely blocked by the second magnet/filter since $\langle S_{-\mathbf{z}}; + | S_{\mathbf{z}}; + \rangle = \langle - | + \rangle = 0$.

2. (b) Now we use

$$p = \cos^2(\pi/4) \cos^2(\pi/4) = 1/4 \quad (28)$$

Same result we obtained in class. This is the famous result and it told us that inserting a middle magnet at 90° scrambled the beam into an equal mixture of spin-up and spin-down along the new axis. Had the middle magnet not been there, we would get $p = 1$.

3. (c) $p = \cos^4(\pi/8) \simeq 0.73$. Again, this case is also odd because splitting a 90° rotation into two 45° steps actually increases the transmission we found in (b). We take advantage of it in the next part.

g) Placing N magnets in the xz -plane at angles $\theta_k = k \frac{\pi/2}{N}$ for $k = 1, \dots, N$, starting from a beam prepared along $\hat{\mathbf{z}}$. Consecutive magnets differ by the fixed step $\Delta\theta = \frac{\pi}{2N}$, so we can apply Eq. (26) for each of the N links to get a transmission of $\cos^2(\Delta\theta/2)$, and

$$p_N = \cos^{2N} \left(\frac{\pi}{4N} \right), \quad (29)$$

as you'd expect. For large N the argument of the cosine is small, so we expand $\cos x \simeq 1 - x^2/2$ to get

$$p_N \simeq \left(1 - \frac{\pi^2}{32N^2} \right)^{2N} \simeq 1 - \frac{\pi^2}{16N} \xrightarrow{N \rightarrow \infty} 1. \quad (30)$$

You can plot this function for large N too if you are not convinced.

After the chain the spin is in $|S_{\mathbf{x}}; +\rangle$, so

$$|\langle S_{\mathbf{x}}; + | S_{\mathbf{x}}; + \rangle|^2 = 1, \quad |\langle S_{\mathbf{x}}; - | S_{\mathbf{x}}; + \rangle|^2 = 0, \quad (31)$$

while measuring along $\hat{\mathbf{z}}$ instead gives, from Eq. (26) with $\theta = 90^\circ$,

$$|\langle S_{\mathbf{z}}; \pm | S_{\mathbf{x}}; + \rangle|^2 = \frac{1}{2} \quad \text{for each sign.} \quad (32)$$

We fully rotate the spin from the $\hat{\mathbf{z}}$ direction to the $\hat{\mathbf{x}}$ direction, as opposed to scrambling it equally between $|S_{\mathbf{x}}; +\rangle$ and $|S_{\mathbf{x}}; -\rangle$ as in (b) above.